

Series and infinite products – 3

Conditional convergence, rearrangements and signs

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Theoretical minimum.

Theorem 1 (Riemann). If $\sum a_n$ converges conditionally, then for every $S \in [-\infty, +\infty]$ there is a permutation σ with $\sum a_{\sigma(n)} = S$; one can also make the partial sums oscillate between any two prescribed bounds. The engine is the pair of identities $\sum a_n^+ = \sum a_n^- = +\infty$ together with $a_n \rightarrow 0$.

Theorem 2 (what a permutation cannot destroy). An absolutely convergent series has the same sum after every rearrangement. For a conditionally convergent one the sum survives under any permutation of bounded displacement, $|\sigma(n) - n| \leq M$, and more generally whenever σ maps each initial segment onto a union of boundedly many intervals.

Theorem 3 (Lévy–Steinitz). For a convergent series of vectors in $\mathbb{R}^d$ the set of sums obtainable by rearrangement is an affine subspace: a point in the absolutely convergent case, and a translate of the annihilator of the set of functionals f with $\sum |f(a_n)| < \infty$ in general.

Fact 1 (signs). For $a_n \downarrow 0$ some choice of signs makes $\sum \varepsilon_n a_n$ converge; convergence for *all* choices means absolute convergence. For random independent signs the dividing line is different: $\sum \pm a_n$ converges almost surely exactly when $\sum a_n^2 < \infty$.

Fact 2 (brackets). Inserting brackets cannot destroy convergence; removing them can. If $a_n \rightarrow 0$ and the brackets have bounded length, the two convergences are equivalent — this is the standard way to prove convergence of a series whose terms cancel in groups.

1. Prove that an absolutely convergent series has the same sum after any rearrangement, and prove Riemann's theorem. Then prove Theorem 2 for permutations with $|\sigma(n) - n| \leq M$ and find where the argument uses the bound.

2. (Putnam 2017) Evaluate the sum

$$\sum_{k=0}^{\infty} \left(\frac{3 \ln(4k+2)}{4k+2} - \frac{\ln(4k+3)}{4k+3} - \frac{\ln(4k+4)}{4k+4} - \frac{\ln(4k+5)}{4k+5} \right).$$

3. (IMC 2000) Does there exist a bijection $\pi: \mathbb{N} \rightarrow \mathbb{N}$ such that the series $\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2}$ is convergent?
4. (ICMC 2024) Let $(t_n)_{n \geq 1}$ be defined by $t_1 = 1$, $t_{2k} = -t_k$ and $t_{2k+1} = t_{k+1}$ for all $k \geq 1$. Consider the series $\sum_{n=1}^{\infty} \frac{t_n}{n^{1/2024}}$. Prove that it converges to a positive real number.
5. Describe the set of all sums $\sum_{n \in A} a_n$, where A runs over subsets of $\mathbb{N}$, for $a_n = \frac{2}{3^n}$, for $a_n = \frac{1}{2^n}$ and for $a_n = \frac{1}{n(n+1)}$. Prove the general rule: the set of subsums is an interval when $a_n \leq \sum_{k > n} a_k$ for all n , and a Cantor set when the opposite strict inequality holds for all n (compare with Topology of the real line – 2).